**St. Francis Institute of Technology, Mumbai-400 103**
**Department Of Information Technology**

**A.Y. 2025-2026**
**Class: BE-ITA/B, Semester: VIII**

Subject: BlockChain Lab

# Experiment 1

1. **Aim:** Case Study on Blockchain

2. **Objectives:** Students should be able to understand application areas of Blockchain Technology in real world.

3. **Prerequisite:** Distributed systems and Security basics

4. **Requirements:** Online/Library resources

5. **Pre-Experiment Exercise:**

6. **Theory:**

## Define Blockchain

Blockchain is a distributed and decentralized digital ledger that records transactions across multiple computers in a secure and immutable manner.
Each transaction is stored in a block, and blocks are linked together using cryptographic hashes, forming a chain. Once data is added, it becomes extremely difficult to modify, ensuring transparency and security.

### Types of Blockchain

| Type | Description |
|---|---|
| Public Blockchain | Open network where anyone can participate and validate transactions. Example: Bitcoin |
| Private Blockchain | Controlled by a single organization with restricted access. |
| Consortium Blockchain | Managed by a group of organizations instead of a single entity. |
| Hybrid Blockchain | Combines features of both public and private blockchains. |

### Features of Blockchain

- **Decentralization**: No central authority controls the network.
- **Transparency** : Transactions are visible to participants.
- **Immutability** : Data once recorded cannot be altered easily.
- **Security** : Uses cryptography to secure data.
- **Consensus Mechanism** : Network nodes agree before transactions are validated.
- **Distributed Ledger** : Data is stored across many nodes.

**Security in Blockchain**

Blockchain ensures security through several mechanisms:

- **Cryptographic Hashing :** Each block has a unique hash linking it to the previous block.
- **Public-Private Key Encryption :** Secures digital signatures.
- **Consensus Algorithms :** Prevent fraudulent transactions.
- **Decentralization :** Reduces risk of single point of failure.
- **Smart Contracts** : Automated secure execution of agreements.

## 7. Laboratory Exercise

### A. Procedure

- Refer to any authenticated web reference to complete the following tables

| S.no | Application Domain | Real Life Application |
|---|---|---|
| 1. | Banking | Faster cross-border payments and fraud prevention using blockchain networks. |
| 2 | Healthcare | Secure storage and sharing of patient medical records across hospitals. |
| 3 | Taxation | Transparent tax collection and tracking government revenue. |
| 4 | Insurance | Automated claim processing through smart contracts. |
| 5 | Gaming | Ownership of in-game assets and NFTs |
| 6 | Mobile Payment | Secure peer-to-peer payments using cryptocurrencies like Bitcoin or Ethereum. |
| 7 | IOT | Secure communication and data exchange between connected devices. |
| 8 | Governance | Transparent electronic voting systems. |
| 9. | Border security | Secure identity verification and tracking of immigration records. |
| 10 | Music | Platforms like Arbit reward musicians fairly for their creative work. |

- Find and Study opensource tools used for blockchain applications.

| S.no | Assignment | Open Source Tool |
|---|---|---|
| 1. | To implement basic hash chain | Python / Node.js cryptographic libraries |
| 2 | Basic Ehereum Transactions | Ethereum with tools like Ganache |
| 3 | Implement Smart Contracts | Remix IDE |
| 4 | Languages used for Blockchain platform | Solidity, JavaScript, Python |

**8. Post-Experiments Exercise:**

**A. Extended Theory:**

- Advantages of Blockchain
- Limitations and Challenges of Blockchain

**9. Conclusion:**

Block chain technology have capability to transform every traditional industry to a new one either it is financial or non-financial .

**10. References:**

[1] Any online /offline reference